ECO310 Midterm Cheat Sheet - Clear Version

1 Core Utility Functions (Know These Cold!)

1.1 Cobb-Douglas: $u = x_1^a x_2^b$

Why it matters: Appears on EVERY exam

The "Magic Rule": Spend a fraction of your income on each good based on the exponent ratios.

- Fraction spent on good 1: $\frac{a}{a+b}$
- Fraction spent on good 2: $\frac{b}{a+b}$

Demand functions:

$$x_1^* = \frac{a}{a+b} \cdot \frac{m}{p_1}, \quad x_2^* = \frac{b}{a+b} \cdot \frac{m}{p_2}$$

Example: $u = x_1^{0.2} x_2^{0.4}$, $m = 15$, $p_1 = 1$, $p_2 = 2$: Normalize $0.2/0.6 = 1/3$. Spend \$5 on good 1 (5 units), \$10 on good 2 (5 units).

MRS: $MRS = -\frac{a}{b} \cdot \frac{x_2}{x_1}$ (slopes of indifference curves)

At optimum: $MRS = -\frac{p_1}{p_2}$ (slope of budget line)

1.2 Perfect Complements: $u = \min\{ax_1, bx_2\}$

Why it matters: Appears on EVERY exam

Key insight: You always consume at the "kink" where $ax_1 = bx_2$ (no point wasting money on extra of one good when you need them together).

Demand functions: Set $ax_1 = bx_2$ and use budget $p_1x_1 + p_2x_2 = m$:

$$x_1^* = \frac{m}{ap_2 + bp_1}, \quad x_2^* = \frac{am}{ap_2 + bp_1}$$

Example: $u = \min\{4x_1, x_2\}$, $m = 20$, $p_1 = p_2 = 1$: Set $4x_1 = x_2$, budget gives $x_1 + 4x_1 = 20 \Rightarrow x_1 = 4$, $x_2 = 16$.

Critical fact for exams: Perfect complements have **ZERO substitution effect** (entire price change is income effect, because you can't substitute between the goods).

1.3 Perfect Substitutes: $u = ax_1 + bx_2$

Why it matters: Tested in 3 out of 4 exams

Key insight: MRS is constant $= -a/b$. You only buy the good with better "bang-per-buck".

Decision rule: Compare $\frac{a}{p_1}$ (utility per dollar of good 1) vs $\frac{b}{p_2}$ (utility per dollar of good 2):

- If $\frac{a}{p_1} > \frac{b}{p_2}$: Buy only good 1, $x_1^* = m/p_1$, $x_2^* = 0$
- If $\frac{a}{p_1} < \frac{b}{p_2}$: Buy only good 2, $x_1^* = 0$, $x_2^* = m/p_2$
- If equal: Any combination on budget line works

Corner solutions: This ALWAYS gives corner solutions unless bang-per-buck is exactly equal.

1.4 CES Utility: $u = (x_1^r + x_2^r)^{1/r}$

Why it matters: Tested in 2025 (and nests all three above!)

Key insight: As parameter changes, CES becomes:

- $r \to -\infty$: Perfect complements
- $r \to 0$: Cobb-Douglas
- $r = 1$: Perfect substitutes

MRS: $MRS = -\left(\frac{x_2}{x_1}\right)^{1-r}$

2 Finding Optimal Bundles (The Process)

2.1 Step 1: Set Up the Problem

Budget constraint: $p_1x_1 + p_2x_2 = m$

- Slope: $-p_1/p_2$
- Intercepts: $(m/p_1, 0)$ and $(0, m/p_2)$

2.2 Step 2: Find MRS

MRS (Marginal Rate of Substitution): The rate at which you're willing to trade good 2 for good 1 while staying on the same indifference curve.

$$MRS = -\frac{dx_2}{dx_1}\Big|_{\text{same utility}} = -\frac{MU_1}{MU_2} = -\frac{\partial u/\partial x_1}{\partial u/\partial x_2}$$

Interpretation: MRS tells you your "personal price ratio" (how you value the goods). At optimum, this must equal the market price ratio.

2.3 Step 3: Optimize (Interior Solution)

Tangency condition: $MRS = -\frac{p_1}{p_2}$

Or equivalently: $\frac{MU_1}{MU_2} = \frac{p_1}{p_2}$ or $\frac{MU_1}{p_1} = \frac{MU_2}{p_2}$

What this means: "Bang-per-buck" must be equal across all goods. If good 1 gave you more utility per dollar, you'd buy more of it.

Budget binds: $p_1x_1 + p_2x_2 = m$ (spend all your money)

Solve these two equations together to get x_1^* and x_2^*.

2.4 Step 4: Check for Corner Solutions

When do corners happen?

- Perfect substitutes (always a corner unless bang-per-buck exactly equal)
- When your FOC (tangency condition) gives you a negative quantity
- When goods aren't worth the price (MRS never equals price ratio)

How to check: If tangency gives $x_i < 0$, set $x_i = 0$ and spend all m on the other good.

3 Marshallian vs Hicksian Demand

3.1 Marshallian Demand: $x_i(p, m)$

What is it: The regular demand function. What you buy given prices and income.

Problem it solves: Maximize utility $u(x_1, x_2)$ subject to budget $p_1x_1 + p_2x_2 = m$

Result: Gives you the best bundle you can afford.

3.2 Hicksian (Compensated) Demand: $h_i(p, \bar{u})$

What is it: The demand that achieves a target utility level at minimum cost.

Problem it solves: Minimize expenditure $p_1x_1 + p_2x_2$ subject to utility constraint $u(x_1, x_2) = \bar{u}$

Result: Gives you the cheapest bundle that achieves utility $\bar{u}$.

When you use it: For substitution effects (holding utility constant while prices change).

3.3 Example: Finding Hicksian Demand

$u = x_1x_2$, target $\bar{u} = 64$, $p_1 = p_2 = 4$:

MRS = price ratio: $x_2/x_1 = 1 \Rightarrow x_1 = x_2$

Use $u = 64$: $x_1^2 = 64 \Rightarrow x_1 = x_2 = 8$

4 Substitution & Income Effects

4.1 The Big Picture

When price of good 1 rises, two things happen:

1. **Substitution effect:** Good 1 is now relatively more expensive, so you substitute away from it (even keeping utility constant)
2. **Income effect:** Your purchasing power decreased, so you adjust consumption based on whether good 1 is normal or inferior

Total effect = Substitution effect + Income effect

4.2 The Recipe (Always Follow These Steps!)

Suppose p_1 increases from p_1^{old} to p_1^{new}:

Step 1: Find **original bundle** (x_1^O, x_2^O) at old prices

Step 2: Find **new bundle** (x_1^N, x_2^N) at new prices

Step 3: Find **compensated bundle** (x_1^C, x_2^C):

- Use new prices p_1^{new}, p_2
- Keep utility at original level: $u(x_1^C, x_2^C) = u(x_1^O, x_2^O)$
- This is the Hicksian demand at new prices!

Step 4: Calculate effects:

- Substitution effect: $\Delta x_1^{\text{sub}} = x_1^C - x_1^O$
- Income effect: $\Delta x_1^{\text{inc}} = x_1^N - x_1^C$
- Total effect: $\Delta x_1^{\text{total}} = x_1^N - x_1^O$

4.3 Sign Rules

Substitution effect: ALWAYS negative (or zero) when price rises

- If p_1 rises, $x_1^C < x_1^O$ (you buy less to substitute away)
- Exception: Perfect complements have zero substitution effect

Income effect: Depends on type of good

- Normal good: Negative (buy less when poorer)
- Inferior good: Positive (buy more when poorer)

4.4 Example with Cobb-Douglas

$u = BT$, $m = 64$, p_B rises from \$1 to \$4, $p_T = \$4$

Original: $B^O = 32$, $T^O = 8$, $u = 256$

New: $B^N = 8$, $T^N = 8$
Compensated: $B = T$ and $BT = 256 \Rightarrow B^C = 16$, $T^C = 16$
Effects: Sub: -16, Income: -8, Total: -24

5 Expected Utility (Key for Every Exam!)

5.1 Setup

Lottery: Get outcome c_i with probability p_i
Expected utility:

$$EU(L) = \sum_i p_i \cdot u(c_i) = p_1 u(c_1) + p_2 u(c_2) + \ldots$$

Critical: Use final wealth in each state, not the change!

5.2 Example: Setting Up EU

Have \$100, invest \$$x$ in risky asset (+100% or -50%, each prob 1/2):
Good: $W = 100 + x$, Bad: $W = 100 - 0.5x$
$EU = \frac{1}{2}u(100 + x) + \frac{1}{2}u(100 - 0.5x)$

5.3 Optimizing EU

To find optimal x:

1. Take derivative: $\frac{d(EU)}{dx}$
2. Set equal to zero: $\frac{d(EU)}{dx} = 0$
3. Solve for x^*
4. Check corners: $x = 0$ or $x =$ max possible

5.4 Risk Aversion

Definition: You prefer certainty to a gamble with the same expected value.
Math version: $u(E[W]) > EU$ (utility of expected wealth > expected utility)
This happens when u is **concave** ($u'' < 0$)
Risk neutral: u is linear, $u(E[W]) = EU$
Risk loving: u is convex ($u'' > 0$), $u(E[W]) < EU$

5.5 Certainty Equivalent (CE)

Definition: The guaranteed amount that gives the same utility as the lottery.

$$u(CE) = EU$$

Risk premium: $RP = E[W] - CE$

- Risk averse: $RP > 0$ (would pay to avoid risk)
- Risk neutral: $RP = 0$

Example: 50-50 of \$0 or \$200, $u(w) = \sqrt{w}$: $EU = \frac{1}{2}\sqrt{200}$, so $CE = 50$. $E[W] = 100$, thus $RP = 50$.

5.6 Common Utility Functions

Log utility: $u(w) = \ln(w)$

- Risk averse (concave)
- Used frequently on exams

CRRA: $u(w) = \frac{w^{1-\rho}}{1-\rho}$ (or $\ln w$ if $\rho = 1$)

- $\rho > 0$: Risk averse
- Higher ρ: More risk averse

6 WARP (Revealed Preference)

6.1 What is WARP?

Weak Axiom of Revealed Preference: If you choose bundle x when bundle y is affordable, then whenever you choose y, bundle x must NOT be affordable (or must cost more).
Interpretation: Your choices should be consistent. If you "revealed" you prefer x to y by choosing it when both were available, you shouldn't later choose y when x is also available.

6.2 How to Check WARP (Two Periods)

Given: Period A bundle (x_1^A, x_2^A), prices (p_1^A, p_2^A), spent m^A
Period B bundle (x_1^B, x_2^B), prices (p_1^B, p_2^B), spent m^B
Step 1: Check if A's bundle was revealed preferred to B's bundle:

- Cost of B's bundle at A's prices: $p_1^A x_1^B + p_2^A x_2^B$
- If this $\leq m^A$, then A's choice revealed preference for A's bundle

Step 2: Check if B's bundle was revealed preferred to A's bundle:

- Cost of A's bundle at B's prices: $p_1^B x_1^A + p_2^B x_2^A$
- If this $\leq m^B$, then B's choice revealed preference for B's bundle

WARP violated if: Both bundles are revealed preferred (at least one strictly)

6.3 Example

Mon: Spent \$90, Tue bundle cost \$80 at Mon prices $\Rightarrow$ Mon strictly revealed preferred
Tue: Spent \$100, Mon bundle cost \$100 at Tue prices $\Rightarrow$ Tue weakly revealed preferred
Result: WARP violated (both revealed preferred)

7 Corner Solutions & Non-Negativity

7.1 When Do Corners Happen?

Situation 1: Perfect substitutes (almost always a corner)
Situation 2: When FOC gives negative quantity

- You set MRS = price ratio, solve, get $x_i < 0$
- This means at any positive quantity, the good isn't worth buying
- Solution: Set $x_i = 0$, spend all income on other good

Situation 3: MRS never equals price ratio

- If $|MRS| > p_1/p_2$ everywhere: consume only good 1
- If $|MRS| < p_1/p_2$ everywhere: consume only good 2

7.2 How to Check in Your Answer

ALWAYS verify: $x_1 \geq 0$ and $x_2 \geq 0$
If either is negative, you made an error or you're at a corner.

8 Quick Tips for Common Exam Traps

1. **Cobb-Douglas:** "Spend 3x as much" $\neq$ "Buy 3x as much" (unless prices are equal)
2. **Perfect complements:** Substitution effect is ALWAYS zero (entire change is income effect)
3. **Portfolio problems:** Variance $\sigma^2 \neq$ Standard deviation σ
4. **EU setup:** Use FINAL wealth in each state, not the change
5. **Marshallian vs Hicksian:** Marshallian maximizes utility given income; Hicksian minimizes cost given utility
6. **Risk aversion:** Check if $u(E[W]) > EU$, not just if u is concave
7. **WARP:** "Revealed preferred" = chosen when other was affordable (cost $\leq$ budget)
8. **Substitution effect:** ALWAYS use compensated bundle (same utility, new prices)